Md Abul Kashem Junaid

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Summary

Worked as an AI Developer for 10 months, building production LLM applications: retrieval-augmented generation (RAG), code intelligence, LLM fine-tuning, and document/identity verification pipelines. Strong in Python, FastAPI, vector databases (Weaviate, LanceDB, ChromaDB, FAISS) and agent frameworks (LangGraph). Fluent in AI-assisted development: uses tools like Claude Code to design, build and ship projects faster without compromising quality. Seeking an AI Engineer role building reliable LLM systems.

Education

North South University, Dhaka — B.Sc. in Computer Science and Engineering | Result: CGPA 3.00 out of 4.00 — Jul 2020 – Mar 2025
Dhaka Imperial College, Dhaka — Higher Secondary School Certificate — 2017 – 2019

Technical Skills

Languages: Python, C++
LLMs & GenAI: RAG, Agentic AI, Multi-Agent Systems, Fine-Tuning (LoRA/QLoRA), Prompt Engineering, HyDE, Cross-Encoder Re-ranking, Hybrid Search, LLM Evaluation
Frameworks & Libraries: PyTorch, Hugging Face Transformers, LangChain, LangGraph, CrewAI, Scikit-learn, SentenceTransformers
Vector DBs & Data: Weaviate, LanceDB, ChromaDB, FAISS, Redis, Pandas, NumPy
Computer Vision / OCR: CLIP, DeepSeek-OCR, Tesseract
Backend & DevOps: FastAPI, Docker, Docker Compose, Git
OS: Linux (Arch, Ubuntu), Windows

Experience

AI-Native Developer — Oct 2026 – Present
NeuronFish, Inc. — *Remote*

- Redesigned the entire frontend of Dikkha AI (SSC 2027 relaunch), an AI study app for Class 9–10 students, using Claude Code and Claude Design.
- Created the new visual identity (mascot, 12 avatars, design system) with Claude Design and rebuilt the Flutter UI on it.
- Built the first-run flow: animated onboarding, student setup (learning-style quiz, study load) and rewards, with accessibility tests.
- Built core features: push notifications, Practice with AI-marked photo answers, timed Model Tests and a streaming AI chat tutor.
- Added the app shell (offline handling, loading states, deep links) and integrated all features, with 1,200+ passing tests.

AI Developer — Sep 2025 – Jun 2026
Ztrios Technologies Ltd. — *Dhaka, Bangladesh*

- Built a production Bengali Physics RAG system with FastAPI, Weaviate and Google Gemini, using hybrid (vector + keyword) search to deliver source-grounded question answering.
- Developed a code intelligence Q&A system using Tree-sitter AST parsing, LanceDB, Redis and LLMs, with HyDE query expansion and cross-encoder re-ranking for semantic search across large repositories; validated on 4 codebases of 100–150 files each.
- Fine-tuned Qwen-family language models on 60M tokens for tax-domain applications; built scalable pipelines for data processing, embedding generation and knowledge retrieval.
- Built an end-to-end Bangladesh NID identity verification pipeline (no database lookup) combining barcode detection, multi-engine barcode decoding, OCR extraction, MRZ parsing and cross-validation, achieving 90% identity matching accuracy.

Projects

Multi-Agent Research Platform — *Python, LangGraph, Groq API, Tavily, DuckDuckGo*

- Built a LangGraph multi-agent system (orchestrator, researcher, critic, analyst) with intent-based routing.
- Added source-grounded web research and a critic layer that detects inconsistencies and assigns confidence scores.

arXiv Agentic RAG Research Platform — *LangGraph, Groq API, ChromaDB, FastAPI, Docker*

- Built an agentic RAG system over arXiv papers with hybrid dense+sparse retrieval, RRF fusion and query decomposition.
- Implemented contradiction detection, consensus classification and citation traceability; deployed as a FastAPI service with Docker Compose.

Document Intelligence for Blurry & Noisy Scans — *Python, DeepSeek-OCR, ChromaDB, Groq API, FastAPI, Streamlit, Docker*

- Built an end-to-end pipeline for OCR extraction, chunking, semantic retrieval and evidence-grounded draft generation.

Image-to-Image RAG for Fashion — *Python, CLIP, FAISS, OpenAI embeddings, PyTorch*

- Built a two-stage fashion search: CLIP indexing with attribute extraction, then FAISS retrieval with metadata-aware re-ranking.

Electrical Wiring Fault Detection with YOLOv8 — *Python, PyTorch, YOLOv8, Ultralytics*

- Trained and tuned YOLOv8 object detectors over four iterations to detect 5 wiring fault classes, reaching 0.89 precision and 0.70 mAP@0.5.

Life Expectancy Prediction — *Python, Scikit-learn, Pandas, Flask*

- Compared six ML models on 2,864 records; Random Forest performed best (R^2 0.996); deployed with Flask.

References

Dr. Mohammad Ashrafuzzaman Khan
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Final-Year Project and Thesis Supervisor

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Professional Supervisor